

On integral-input-to-state stabilization*

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Abstract

This paper continues the investigation of the recently introduced integral version of input-to-state stability (iISS). We study the problem of designing control laws that achieve iISS disturbance attenuation. The main contribution is an appropriate concept of control Lyapunov function (iISS-CLF), whose existence leads to an explicit construction of such a control law. The results are compared and contrasted with the ones available for the ISS case.

1 Introduction

Since the concept of input-to-state stability (ISS) was first introduced in [11], there has been a great deal of research on the problem of designing input-to-state stabilizing controllers [4, 6, 7, 9, 15, 16, 17]. In its usual setting, this problem consists in finding a control law that makes the closed-loop system input-to-state stable with respect to external disturbances. Most of this activity has centered around the concept of ISS-control Lyapunov function (ISS-CLF). The existence of an ISS-CLF has been shown to be necessary and sufficient for the existence of an input-to-state stabilizing state feedback control law. In fact, the knowledge of an ISS-CLF leads to explicit formulae for such control laws (see [4] and [15, 17] for two different constructions). For certain classes of systems, ISS-CLFs can be systematically generated via backstepping [7, 9]. In addition, input-to-state stabilizing control laws possess desirable properties associated with inverse optimality [4, 9].

In parallel with these developments, an integral variant of input-to-state stability (iISS) has been introduced and studied in [1, 12]. Intuitively, while the state of an input-to-state stable system is

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small if inputs are small (cf. “ L^∞ to L^∞ stability”), the state of an integral-input-to-state stable system is small if inputs have finite energy as defined by an appropriate integral (cf., e.g., “ L^2 to L^∞ stability”). The concept of iISS is weaker than that of ISS, in the sense that every input-to-state stable system is necessarily integral-input-to-state stable but the converse is not true. From the viewpoint of control design for systems with disturbances, this leads to the existence of systems that are integral-input-to-state stabilizable but not input-to-state stabilizable (an example is given below). This paper is concerned with the problem of designing integral-input-to-state stabilizing control laws, which apparently has not been studied in the literature. We introduce the concept of iISS-CLF, whose existence leads to an explicit construction of an integral-input-to-state stabilizing state feedback control law. Although the results we obtain are similar to those available for the ISS case, some nontrivial modifications of existing notions and techniques are required. A new equivalent characterization of iISS is also established. These developments are heavily based on the work by David Angeli and two of the authors reported in [1], and the characterization given here is actually a minor variation of one of the properties that were established in that paper. Many issues that are resolved for ISS, some of which we mentioned at the beginning, remain to be investigated in the iISS case.

The notion of iISS is precisely defined in the next section. For the definition of ISS the reader is referred to [11, 6, 7, 13]. The concept of ISS is known to be useful in the context of control of uncertain nonlinear systems. In particular, the problem of adaptive nonlinear stabilization can be reduced, via the modular approach proposed in [8], to the problem of input-to-state stabilization with respect to the parameter estimation error. Similar techniques are exploited in the nonlinear supervisory control setting presented in [5], where input-to-state stability is used to prove that the closed-loop system is detectable through the estimation error, which leads to systematic hybrid output feedback stabilization methods. The applicability of this latter idea in the iISS framework will be discussed in a future publication.

2 Characterizations of iISS

Consider the system

$$\dot{x} = f(x, d), \quad x \in \mathbb{R}^n, \quad d \in \mathbb{R}^k \quad (1)$$

where $f : \mathbb{R}^n \times \mathbb{R}^k \rightarrow \mathbb{R}^n$ is a locally Lipschitz function and d can be viewed either as a disturbance or as a control input. Following [12], we will call the system (1) *integral-input-to-state stable (iISS)* with respect to d if for some functions $\alpha, \gamma \in \mathcal{K}_\infty$ and $\beta \in \mathcal{KL}$, for all initial states x_0 , and all $d(\cdot)$ the following estimate holds:

$$\alpha(|x(t)|) \leq \beta(|x_0|, t) + \int_0^t \gamma(|d(s)|) ds \quad \forall t \geq 0.$$

It was shown in [1] that (1) is iISS with respect to d if and only if there exists an *iISS-Lyapunov function*, i.e., a positive definite proper smooth function $V : \mathbb{R}^n \rightarrow \mathbb{R}$ such that for some class $\mathcal{K}$ function σ and some positive definite function α we have

$$\nabla V(x) f(x, d) \leq -\alpha(|x|) + \sigma(|d|) \quad \forall x, d. \quad (2)$$

Two other necessary and sufficient conditions for iISS were given in [1]. One of them states that (1) is iISS if and only if it is *θ -GAS* (i.e., the system $\dot{x} = f(x, 0)$ is globally asymptotically stable)

and *zero-output dissipative*, i.e., there exist a positive definite proper smooth¹ function $V : \mathbb{R}^n \rightarrow \mathbb{R}$ and a class $\mathcal{K}$ function σ such that

$$\nabla V(x)f(x, d) \leq \sigma(|d|) \quad \forall x, d. \quad (3)$$

A new equivalent characterization of iISS is established below. This characterization, as well as the ones mentioned above, will be useful in studying the concept of iISS-CLF (to be introduced in Section 3).

Proposition 1 *The following statements are equivalent:*

- (i) *The system (1) is iISS.*
- (ii) *There exist a positive definite proper smooth function $W : \mathbb{R}^n \rightarrow \mathbb{R}$, two class $\mathcal{K}_\infty$ functions ρ and γ , and a positive definite function b with $\int_0^{+\infty} \frac{1}{1+b(r)} dr = +\infty$ such that for all $x \neq 0$ and all d we have*

$$|x| \geq \rho(|d|) \Rightarrow \nabla W(x)f(x, d) < \gamma(|d|)b(W(x)). \quad (4)$$

- (iii) *There exist a positive definite proper smooth function $W : \mathbb{R}^n \rightarrow \mathbb{R}$, two class $\mathcal{K}_\infty$ functions ρ and γ , a positive definite function α , and a positive definite function b with $\int_0^{+\infty} \frac{1}{1+b(r)} dr = +\infty$ such that for all x and d we have*

$$|x| \geq \rho(|d|) \Rightarrow \nabla W(x)f(x, d) \leq -\alpha(|x|) + \gamma(|d|)b(W(x)). \quad (5)$$

PROOF. The implication (iii) $\Rightarrow$ (ii) is trivial. To prove (ii) $\Rightarrow$ (i), suppose that (4) holds. According to [1, Theorem 1], we need to verify that (1) is 0-GAS and zero-output dissipative. The 0-GAS property is satisfied because $\nabla W(x)f(x, 0) < 0 \quad \forall x \neq 0$, so that W is a Lyapunov function for the system $\dot{x} = f(x, 0)$. To establish zero-output dissipativity, we define the function

$$V(x) := \int_0^{W(x)} \frac{1}{1+b(r)} dr$$

which is clearly proper and positive definite by hypotheses. When $|x| \geq \rho(|d|)$ we have

$$\nabla V(x)f(x, d) = \frac{\nabla W(x)f(x, d)}{1+b(W(x))} \leq \frac{\gamma(|d|)b(W(x))}{1+b(W(x))} \leq \gamma(|d|)$$

while when $|x| < \rho(|d|)$ we have

$$\nabla V(x)f(x, d) \leq \max_{|x| \leq \rho(|d|)} \nabla V(x)f(x, d) =: \hat{\gamma}(|d|).$$

Defining $\sigma(r) := \max\{\gamma(r), \hat{\gamma}(r)\}$ for each $r \geq 0$, we obtain (3) as needed.

It remains to prove (i) $\Rightarrow$ (iii). Suppose that (1) is iISS. By [1, Theorem 1], there exists an iISS-Lyapunov function V so that (2) holds. Define $W(x) := e^{V(x)} - 1$. This function is proper positive definite because V is, thus there exists a function $\hat{\alpha} \in \mathcal{K}_\infty$ such that

$$\hat{\alpha}(|x|) \leq W(x) \quad \forall x. \quad (6)$$

¹The smoothness assumption can actually be weakened, cf. Remark 2.3 in [1]

Define a function $\rho \in \mathcal{K}$ by $\rho(r) := \hat{\alpha}^{-1}(\sqrt{\sigma(r)})$, $r \geq 0$. Whenever $|x| \geq \rho(|d|)$, from (6) we obtain $W(x) \geq \hat{\alpha}(|x|) \geq \sqrt{\sigma(|d|)}$. Therefore, using (2) we have

$$\begin{aligned}\nabla W(x)f(x, d) &= \nabla V(x)f(x, d)e^{V(x)} = \nabla V(x)f(x, d)(W(x) + 1) \\ &\leq -\alpha(|x|)(W(x) + 1) + \sigma(|d|)(W(x) + 1) \\ &\leq -\alpha(|x|) + \sigma(|d|)W(x) + \sigma(|d|) \leq -\alpha(|x|) + \sigma(|d|)W(x) + \sqrt{\sigma(|d|)}W(x).\end{aligned}$$

Thus (5) is satisfied with $\gamma(r) := \sigma(r) + \sqrt{\sigma(r)}$ and $b(r) := r$. $\square$

We know from [12, Theorem 2] that if there exist a positive definite proper smooth function $W : \mathbb{R}^n \rightarrow \mathbb{R}$, a constant $q > 0$, and two class $\mathcal{K}_\infty$ functions γ and χ such that

$$\nabla W(x)f(x, d) \leq (\gamma(|d|) - q)W(x) + \chi(|d|) \quad \forall x, d$$

then (1) is iISS. It is also remarked in [12] that, contrary to what the corresponding developments for ISS might suggest, merely asking that

$$\nabla W(x)f(x, d) \leq (\gamma(|d|) - q)\alpha(|x|) + \chi(|d|) \quad \forall x, d$$

for some $\alpha, \gamma, \chi \in \mathcal{K}_\infty$ is *not* sufficient for iISS. A counterexample given there is $\dot{x} = (d^2 - 1)x^3$. Taking $W(x) := x^2$, we obtain $\nabla W(x)f(x, d) = (d^2 - 1)2x^4$, but solutions may escape to infinity in finite time. It follows from Proposition 1 that a suitable “compromise” is to impose the condition

$$\nabla W(x)f(x, d) \leq \gamma(|d|)W(x) - q\alpha(|x|) + \chi(|d|) \quad \forall x, d \quad (7)$$

with $q > 0$ and $\alpha, \gamma, \chi \in \mathcal{K}_\infty$, as expressed in the following result.

Corollary 2 *If there exist a positive definite proper smooth function $W : \mathbb{R}^n \rightarrow \mathbb{R}$, a constant $q > 0$, and class $\mathcal{K}_\infty$ functions α , γ and χ such that (7) is satisfied, then the system (1) is iISS.*

PROOF. It is straightforward to check that (4) holds with $\rho(r) := \alpha^{-1}(2\chi(r)/q)$ and $b(r) := r$. $\square$

3 Control Lyapunov functions

Consider the system

$$\dot{x} = f(x) + G_1(x)d + G_2(x)u, \quad x \in \mathbb{R}^n, \quad d \in \mathbb{R}^k, \quad u \in \mathbb{R}^m \quad (8)$$

where $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$, $G_1 : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times k}$ and $G_2 : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times m}$ are locally Lipschitz functions. We will be interested in the design of integral-input-to-state stabilizing feedback control laws for (8), i.e., control laws of the form $u = k(x, d)$ or $u = k(x)$ such that the closed-loop system is iISS with respect to the disturbance d . We first introduce an appropriate notion of control Lyapunov function. Given a positive definite proper smooth function $V : \mathbb{R}^n \rightarrow \mathbb{R}$, we will call it an *iISS-CLF of the 1st type* for the system (8) if there exists a class $\mathcal{K}_\infty$ function σ such that for all $x \neq 0$ and all d we have

$$\inf_u \{L_f V(x) + L_{G_1} V(x)d + L_{G_2} V(x)u - \sigma(|d|)\} < 0. \quad (9)$$

For each x here $L_f V(x) := \nabla V(x)f(x)$ is a scalar and $L_{G_i} V(x) := \nabla V(x)G_i(x)$ ($i = 1, 2$) are row vectors. The inequality (9) can be rewritten as

$$L_{G_2} V(x) = 0 \Rightarrow L_f V(x) + L_{G_1} V(x)d - \sigma(|d|) < 0. \quad (10)$$

Let us define the set $D_0 := \{x \in \mathbb{R}^n : L_{G_2} V(x) = 0\}$. The following lemma recasts the definition of an iISS-CLF of the 1st type in a more concise way.

Lemma 3 *A positive definite proper smooth function V is an iISS-CLF of the 1st type for (8) if and only if there exist a positive definite function α and a class $\mathcal{K}$ function χ such that for all x and d we have*

$$L_{G_2} V(x) = 0 \Rightarrow L_f V(x) + L_{G_1} V(x)d \leq -\alpha(|x|) + \chi(|d|). \quad (11)$$

PROOF. It is clear that (11) implies (9) with $\sigma(r) = \chi(r)$. Now assume that (9) holds. Then we have $L_f V(x) + L_{G_1} V(x)d - \sigma(|d|) < 0$ for all $x \in D_0 \setminus \{0\}$ and all d . Let α be any positive definite continuous function such that for all $r \geq 0$, all $x \in D_0$ with $|x| = r$, and all $|d| \leq r$ we have

$$\alpha(r) \leq -L_f V(x) - L_{G_1} V(x)d + \sigma(|d|).$$

Such a function α exists because for any two numbers $0 < r < R$ there exists a constant $c > 0$ with the property that $-L_f V(x) - L_{G_1} V(x)d + \sigma(|d|) > c$ for all $(x, d) \in S_{r,R} := \{(x, d) : x \in D_0, r \leq |x| \leq R, |d| \leq R\}$. Indeed, if $S_{r,R}$ is empty, any $c > 0$ can be used, and if $S_{r,R}$ is nonempty, the continuous function $-L_f V(x) - L_{G_1} V(x)d + \sigma(|d|)$ is bounded away from zero. Thus, if $x \in D_0 \setminus \{0\}$ and $|x| \geq |d|$, we have

$$L_f V(x) + L_{G_1} V(x)d - \sigma(|d|) \leq -\alpha(|x|). \quad (12)$$

Next, let $\hat{\chi}$ be any class $\mathcal{K}$ function with the property that

$$\hat{\chi}(r) \geq L_f V(x) + L_{G_1} V(x)d - \sigma(|d|) + \alpha(|x|)$$

for all $r \geq 0$, all $x \in D_0$ with $|x| \leq r$, and all $|d| = r$. (Arguing as above, one can show that such a function $\hat{\chi}$ exists because for any two numbers $0 < r < R$ there exists a constant $c > 0$ with the property that $-L_f V(x) - L_{G_1} V(x)d + \sigma(|d|) < c$ for all $(x, d) \in S_{r,R}$. Note that such an upper bound c can be taken to be small if R is small.) Therefore, if $x \in D_0 \setminus \{0\}$ and $|x| < |d|$, we have

$$L_f V(x) + L_{G_1} V(x)d - \sigma(|d|) \leq -\alpha(|x|) + \hat{\chi}(|d|).$$

Together with (12), this implies that (11) holds with $\chi(r) := \hat{\chi}(r) + \sigma(r)$. $\square$

The link between integral-input-to-state stabilization and the existence of an iISS-CLF of the 1st type is provided by the following result.

Proposition 4 *If the system (8) admits an iISS-CLF of the 1st type, then there exists an integral-input-to-state stabilizing feedback $u = k(x, d)$ that is smooth when $x \neq 0$. Conversely, if there exists an integral-input-to-state stabilizing feedback $u = k(x, d)$ that is locally Lipschitz everywhere, then (8) admits an iISS-CLF of the 1st type.*

PROOF. If there exists an integral-input-to-state stabilizing feedback $u = k(x, d)$ that is locally Lipschitz everywhere, then the closed-loop system

$$\dot{x} = f(x) + G_1(x)d + G_2(x)k(x, d) \quad (13)$$

satisfies the assumptions of Theorem 1 in [1]. This implies that (13) admits an iISS-Lyapunov function, which is then automatically an iISS-CLF of the 1st type for (8).

Now suppose that V is an iISS-CLF of the 1st type for (8). The “universal” formula given in [10] (see also [13]) yields the feedback control law

$$k(x, d) := \begin{cases} -\frac{\omega + \sqrt{\omega^2 + (L_{G_2}V(x)(L_{G_2}V(x))^T)^2}}{L_{G_2}V(x)(L_{G_2}V(x))^T}(L_{G_2}V(x))^T, & L_{G_2}V(x) \neq 0 \\ 0, & L_{G_2}V(x) = 0 \end{cases}$$

where $\omega := L_fV(x) + L_{G_1}V(x)d - \sigma(|d|)$. It follows from [10, 13] that this control is smooth when $x \neq 0$. Indeed, the mapping $(x, d) \mapsto (\omega, L_{G_2}V(x)(L_{G_2}V(x))^T)$ sends each pair (x, d) with $x \neq 0$ into the set $S := \{(a, b) \in \mathbb{R}^2 : b > 0 \text{ or } a < 0\}$ on which the mapping $(a, b) \mapsto -\frac{a + \sqrt{a^2 + b^2}}{b}$ is analytic, and k is defined via the composition of these two mappings. Along the solutions of (13) we have

$$\dot{V} - \sigma(|d|) = -\sqrt{\omega^2 + (L_{G_2}V(x)(L_{G_2}V(x))^T)^2}$$

and this is negative if $x \neq 0$ by virtue of (10). Thus the system (13) is 0-GAS and zero-output dissipative, hence iISS. $\square$

Motivated by the developments of Section 2, we can introduce an alternative definition of iISS-CLF as follows. Given a positive definite proper smooth function $W : \mathbb{R}^n \rightarrow \mathbb{R}$, we will call it an *iISS-CLF of the 2nd type* for the system (8) if there exist two class $\mathcal{K}_\infty$ functions ρ and γ and a positive definite function b with $\int_0^{+\infty} \frac{1}{1+b(r)} dr = +\infty$ such that for all $x \neq 0$ and all d we have

$$\begin{aligned} |x| &\geq \rho(|d|) \\ &\Downarrow \\ \inf_u \{L_fW(x) + L_{G_1}W(x)d + L_{G_2}W(x)u - \gamma(|d|)b(W(x))\} &< 0 \end{aligned}$$

We then have the following result.

Proposition 5 *If the system (8) admits an iISS-CLF of the 2nd type, then there exists an integral-input-to-state stabilizing feedback $u = k(x, d)$ that is smooth when $(x, d) \neq (0, 0)$. Conversely, if there exists an integral-input-to-state stabilizing feedback $u = k(x, d)$ that is locally Lipschitz everywhere, then (8) admits an iISS-CLF of the 2nd type.*

PROOF. If there exists an integral-input-to-state stabilizing feedback $u = k(x, d)$ that is locally Lipschitz everywhere, then the closed-loop system (13) satisfies the assumptions of Proposition 1. This implies that there exists a function W that satisfies (4), and we see that W is automatically an iISS-CLF of the 2nd type for (8).

Now suppose that W is an iISS-CLF of the 2nd type for (8). The proof of Proposition 4 would suggest the control law

$$\hat{k}(x, d) := \begin{cases} -\frac{\omega + \sqrt{\omega^2 + (L_{G_2}W(x)(L_{G_2}W(x))^T)^2}}{L_{G_2}W(x)(L_{G_2}W(x))^T}(L_{G_2}W(x))^T, & L_{G_2}W(x) \neq 0 \\ 0, & L_{G_2}W(x) = 0 \end{cases}$$

where $\omega := L_f W(x) + L_{G_1} W(x)d - \gamma(|d|)b(W(x))$. However, this control law may fail to be well defined on the set $\{(x, d) : L_{G_2} W(x) = 0, |x| < \rho(|d|)\}$ since ω is not necessarily negative there. To fix this problem, we modify the control law as follows. We start by finding a smooth function $\varphi : \mathbb{R}^n \times \mathbb{R}^k \setminus \{(0, 0)\} \rightarrow [0, 1]$ such that $\varphi(x, d) = 0$ if $|x| \leq \rho(|d|)$ and $\varphi(x, d) = 1$ if $|x| \geq 2\rho(|d|)$. It is well known that such a “bump” function exists because $S_1 := \{(x, d) : |x| \leq \rho(|d|)\}$ and $S_2 := \{(x, d) : |x| \geq 2\rho(|d|)\}$ are disjoint closed subsets of the manifold $\mathbb{R}^n \times \mathbb{R}^k \setminus \{(0, 0)\}$ (see, e.g., [3, Lemma 3.1.2]). Then we let

$$k(x, d) := \begin{cases} -\frac{\omega + \sqrt{\omega^2 + (L_{G_2} W(x)(L_{G_2} W(x))^T)^2}}{L_{G_2} W(x)(L_{G_2} W(x))^T} (L_{G_2} W(x))^T \varphi(x, d), & L_{G_2} W(x) \neq 0 \\ 0, & L_{G_2} W(x) = 0 \end{cases}$$

(note that $L_{G_2} W(0) = 0$, so the fact that $\varphi(0, 0)$ is undefined does not matter.) Since on the set $S_3 := \{(x, d) : |x| \geq \rho(|d|), (x, d) \neq (0, 0)\}$ we have $\omega < 0$ whenever $L_{G_2} W(x) = 0$, there is a neighborhood D of S_3 such that $\omega < 0$ whenever $L_{G_2} W(x) = 0$ and $(x, d) \in D$. This implies that the function $\hat{k}$ is smooth on D , from which it follows that k is smooth on D . Also note that k is smooth on the interior $\mathring{S}_1$ of S_1 because $k \equiv 0$ on $S_1 \setminus \{(0, 0)\}$. Since $\mathbb{R}^n \times \mathbb{R}^k \setminus \{(0, 0)\} = D \cup \mathring{S}_1$, we conclude that k is smooth on $\mathbb{R}^n \times \mathbb{R}^k \setminus \{(0, 0)\}$. Along the solutions of (13) we have

$$|x| \geq 2\rho(|d|) \Rightarrow \dot{W} - \gamma(|d|)b(W(x)) = -\sqrt{\omega^2 + (L_{G_2} W(x)(L_{G_2} W(x))^T)^2}$$

and this is negative if $x \neq 0$. Thus the system (13) is iISS by virtue of Proposition 1. $\square$

The above results can be improved in two different ways. First, under certain additional assumptions on iISS-CLFs the control laws that we have constructed can be shown to be continuous everywhere. Second, the requirement that an integral-input-to-state stabilizing control law be Lipschitz everywhere is not necessary for the existence of an iISS-CLF. These observations are subsumed in the discussion that will be given at the end of Section 4.

It is not hard to see from the proof of Proposition 1 and from the definitions of iISS-CLFs that if V is an iISS-CLF of the 1st type, then an iISS-CLF of the 2nd type (with $b(r) = r$) can be defined by $W(x) := e^{V(x)} - 1$. On the other hand, if W is an iISS-CLF of the 2nd type, then an iISS-CLF of the 1st type can be defined by $V(x) := \int_0^{W(x)} \frac{1}{1+b(r)} dr$. For example, $V(x) = \ln(1 + W(x))$ if $b(r) = r$.

Recall that a positive definite proper smooth function $W : \mathbb{R}^n \rightarrow \mathbb{R}$ is an *iISS-CLF* for (8) if there exists a class $\mathcal{K}_\infty$ function ρ such that for all $x \neq 0$ and all d we have

$$\begin{aligned} |x| \geq \rho(|d|) \\ \Downarrow \\ \inf_u \{L_f W(x) + L_{G_1} W(x)d + L_{G_2} W(x)u\} < 0 \end{aligned} \tag{14}$$

(see, e.g., [6, 9]). Any ISS-CLF is automatically an iISS-CLF of the 2nd type as can be readily seen from the above definitions, but the converse is not true. In fact, there are systems for which an iISS-CLF can be found while no ISS-CLF exists (see the example below), so the concept of iISS-CLF is indeed a meaningful one.

EXAMPLE. Consider the two-dimensional system

$$\begin{aligned} \dot{x} &= -x + xd + u - x^2 d \\ \dot{y} &= -y + yd - u + x^2 d \end{aligned} \tag{15}$$

No matter what control is applied, $d \equiv 2$ gives $\frac{d}{dt}(x + y) = x + y$. This means that a bounded disturbance may lead to an unbounded trajectory, and hence the system (15) is not input-to-state stabilizable (and thus does not admit an ISS-CLF). On the other hand, it is not hard to see that it is integral-input-to-state stabilizable: setting $u = x^2 d$, we obtain the system

$$\begin{aligned}\dot{x} &= -x + xd \\ \dot{y} &= -y + yd\end{aligned}\tag{16}$$

which is known to be iISS [12]. As an iISS-CLF one can take $W = x^2 + y^2$. Along the solutions of (16) we have $\dot{W} = -2W + 2Wd < \gamma(|d|)W \ \forall x \neq 0$, where $\gamma(r) := 2r$, hence W is an iISS-CLF of the 2nd type (with $b(r) = r$). It is not obvious how to integral-input-to-state stabilize the system (15) without canceling some of the terms that contain the disturbance. However, we will show in the next section that if there exists an iISS-CLF, then we can always find an integral-input-to-state stabilizing control in the pure state feedback form, i.e., one that does not depend on d .

4 State feedback

We showed in the previous section how an integral-input-to-state stabilizing feedback control of the form $u = k(x, d)$ can be constructed from an iISS-CLF. This result is useful when the disturbance d can be directly measured and used for control. The next logical step is to ask whether the existence of an iISS-CLF implies that an integral-input-to-state stabilizing control in the state feedback form $u = k(x)$ can be found. In this section we give a positive answer to this question. Regarding the ISS case, it is shown, e.g., in [9] that (14) is equivalent to the condition that for all $x \neq 0$ we have

$$L_{G_2}V(x) = 0 \Rightarrow L_fV(x) + |L_{G_1}V(x)|\rho^{-1}(|x|) < 0$$

and thus the “universal” construction of the type described in [10, 13] yields an input-to-state stabilizing control law that depends only on the state x and not on the disturbance d . It seems that in the iISS case a more involved argument is required.

Proposition 6 *If the system (8) admits an iISS-CLF of the 1st type, then there exists an integral-input-to-state stabilizing feedback $u = k(x)$ that is smooth on $\mathbb{R}^n \setminus \{0\}$.*

PROOF. Suppose that V is an iISS-CLF of the 1st type. Let $a(x) := L_fV(x)$, $b_1(x) := L_{G_1}V(x)$, and $b_2(x) := L_{G_2}V(x)$. By Lemma 3, there exist a positive definite function α and a class $\mathcal{K}$ function χ such that for all x and d we have

$$b_2(x) = 0 \Rightarrow a(x) + b_1(x)d \leq -\alpha(|x|) + \chi(|d|).$$

In particular, letting $d = 0$ we obtain

$$b_2(x) = 0 \Rightarrow a(x) < 0 \quad \forall x \neq 0.\tag{17}$$

This means that V is a CLF (in the sense of [10, 13]) for the system $\dot{x} = f(x) + G_2(x)u$. Recall that D_0 was defined to be the set $\{x \in \mathbb{R}^n : b_2(x) = 0\}$. Let D_1 be a neighborhood of $D_0 \setminus \{0\}$ in

$\mathbb{R}^n$ such that for each $x \in D_1$ there exists some $x_0 \in D_0 \setminus \{0\}$ with $|a(x) - a(x_0)| < \alpha(|x_0|)$ and $|b_1(x) - b_1(x_0)| < 1$. Then for each $x \in D_1$ and each d we have (picking an appropriate x_0 for x):

$$\begin{aligned} a(x) + b_1(x)d &= a(x_0) + b_1(x_0)d + (a(x) - a(x_0)) + (b_1(x) - b_1(x_0))d \\ &< -\alpha(|x_0|) + \chi(|d|) + \alpha(|x_0|) + |d| = \hat{\sigma}(|d|) \end{aligned}$$

where $\hat{\sigma}(r) := \chi(r) + r$. Let D_2 be a neighborhood of 0 in $\mathbb{R}^n$ such that $|L_{G_1}V(x)| \leq 1$ for all $x \in D_2$. Then for all $x \in D_2$ and all d we have

$$a(x) + b_1(x)d + b_2(x)u \leq a(x) + |d| + b_2(x)u.$$

Note that $D_1 \cup D_2$ is a neighborhood of D_0 in $\mathbb{R}^n$. Let $\varphi(x) : \mathbb{R}^n \rightarrow [0, 1]$ be a smooth function such that $\varphi(x) = 0$ if $x \in D_0$ and $\varphi(x) = 1$ if $x \notin D_1 \cup D_2$. Such a “bump” function is well known to exist (see, e.g., [3, Lemma 3.1.2]). Consider the feedback law of the form $k(x) = k_0(x) + k_1(x)$, where

$$k_0(x) := \begin{cases} -\frac{a(x) + \sqrt{(a(x))^2 + (b_2(x)b_2(x)^T)^2}}{b_2(x)b_2(x)^T} b_2(x)^T, & b_2(x) \neq 0 \\ 0, & b_2(x) = 0 \end{cases}$$

and

$$k_1(x) := \begin{cases} -\frac{b_1(x)b_1(x)^T}{b_2(x)b_2(x)^T} b_2(x)^T \varphi(x), & b_2(x) \neq 0 \\ 0, & b_2(x) = 0 \end{cases}$$

Observe that $b_2(x)k_0(x) \leq 0$ and $b_2(x)k_1(x) \leq 0$ for all x , so also $b_2(x)k(x) \leq 0$ for all x . For all (x, d) with $x \in D_1$ we have

$$a(x) + b_1(x)d + b_2(x)k(x) < \hat{\sigma}(|d|) + b_2(x)k(x) \leq \hat{\sigma}(|d|).$$

For all $(x, d) \neq (0, 0)$ with $x \in D_2$ we have

$$a(x) + b_1(x)d + b_2(x)k(x) \leq -\sqrt{(a(x))^2 + |b_2(x)|^4} + |d| + b_2(x)k_1(x) < 2|d|$$

(because either $b_2(x) = 0$, in which case $a(x) < 0$ by (17), or $b_2(x) \neq 0$, in which case $a(x) + b_2(x)k_0(x) < 0$ by definition of k_0). Finally, for all (x, d) with $x \notin D_1 \cup D_2$ we have

$$\begin{aligned} a(x) + b_1(x)d + b_2(x)k(x) &\leq -\sqrt{(a(x))^2 + |b_2(x)|^4} + b_1(x)d - |b_1(x)|^2 \\ &\leq -\sqrt{(a(x))^2 + |b_2(x)|^4} + |d|^2/4 < |d|^2/2. \end{aligned}$$

Putting the above inequalities together, we obtain

$$a(x) + b_1(x)d + b_2(x)k(x) < \sigma(|d|) \quad \forall (x, d) \neq (0, 0)$$

where $\sigma(r) := \max\{\hat{\sigma}(r), r^2/2, 2r\}$. This implies that the closed-loop system is 0-GAS and zero-output dissipative, hence iISS. Finally, to see that $k(x)$ is smooth on $\mathbb{R}^n \setminus \{0\}$, notice that $k_1(x)$ is smooth everywhere while $k_0(x)$ is smooth on $\mathbb{R}^n \setminus \{0\}$. $\square$

We know from [10, 13] that the control law $k(x)$ defined in the above proof will in addition be continuous at $x = 0$ if V satisfies the following *small control property*: for each $\epsilon > 0$ there exists a $\delta > 0$ such that whenever $0 < |x| < \delta$ there exists some u with $|u| < \epsilon$ for which

$$L_f V(x) + L_{G_2} V(x)u < 0. \tag{18}$$

One situation in which this happens is when there exists an integral-input-to-state stabilizing feedback $u(k, d)$, locally Lipschitz everywhere, such that $k(0, 0) = 0$ (as in the example of Section 3). Indeed, in this case there exists an iISS-CLF of the 1st type V that satisfies (9) for some $\sigma \in \mathcal{K}_\infty$. Letting $d = 0$ gives $L_f V(x) + L_{G_2} V(x)k(x, 0) < 0$. Since $k(x, d)$ is continuous and $k(0, 0) = 0$, we see that the small control property is satisfied.

Note that we have not yet obtained a necessary and sufficient condition for integral-input-to-state stabilization, since the necessity parts of Propositions 4 and 5 require that the control be locally Lipschitz at $x = 0$. A similar issue in fact arises in the context of ISS, and it has been frequently overlooked. One way to fix this problem in the case of an input-to-state stabilizing control law that is only smooth when $x \neq 0$ is to consider a positive definite function α such that the right-hand side of the closed-loop system becomes smooth everywhere when multiplied by $\alpha(|x|)$. Then the results of [14] allow one to conclude that this modified system will possess an ISS-CLF, which will in turn be an ISS-CLF for the original system because α is positive definite (a similar trick was used in [2], although for a different purpose, namely, assuring continuity at the origin of a possibly discontinuous vector field). Alternatively, one can appeal to the results of [17] where a generalization of a converse Lyapunov theorem was proved for systems that do not have the regularity property on the invariant sets. As for the iISS case, a careful examination of the argument in [1] reveals that it does not use the assumption that the right-hand side is Lipschitz at zero. Therefore, to ensure the existence of an iISS-CLF is enough to require that an integral-input-to-state stabilizing state feedback be smooth when $x \neq 0$ and continuous at 0. Since the existence of an iISS-CLF of the 2nd type implies the existence of an iISS-CLF of the 1st type, we can summarize our main findings as follows.

Theorem 7 *The system (8) admits an integral-input-to-state stabilizing state feedback control law $u = k(x)$ that is smooth on $\mathbb{R}^n \setminus \{0\}$ and continuous at 0 if and only if there exists an iISS-CLF of either the 1st or the 2nd type that satisfies the small control property.*

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